E.2 P_{regret}-TRAINED condition Interface details

We progressively teach subjects how to compute the regret of a trajectory segment by sequentially introducing its components outlined in Equation 3. When subjects interact with episodes from the delivery domain, we display the components of regret as we introduce them. First, we introduce the "score so far," $\Sigma_\sigma r$. Next, we explain the "biggest possible score increase," $V_r^*(s^\sigma_{|\sigma|})$. Finally, we present the "biggest possible final score," which combines both components as $\Sigma_\sigma r + V_r^*(s^\sigma_{|\sigma|})$. For each component, we explain how it is computed and allow subjects to interact with various tasks to observe the corresponding values. Additionally, we ask subjects to compute $V_r^*(s^\sigma_{|\sigma|})$ and $\Sigma_\sigma r + V_r^*(s^\sigma_{|\sigma|})$ for three different trajectory segments each, providing feedback on their answers. Since we always present segment pairs that share the same start state, we do not introduce $V_r^*(s^\sigma_0)$ to subjects because this component cancels out when computing preference distributions using the regret preference model (see Appendix B.2).

After understanding how to compute regret, subjects are taught to use it when generating preferences following a procedure similar to that in Section E.1. They are first asked to label preferences for six trajectory segments. Then, they are instructed to generate preference labels based on the "biggest possible final score" and shown a detailed example. Subjects label six more trajectory segments, receiving feedback on the correctness of their preferences as illustrated in Figure 18. Following this, they are instructed on how *not* to generate preferences (e.g., "do not select the path that merely has the higher score so far"). Finally, they are given another six segment pairs to label, with feedback provided on their preferences. Note that these pedagogical segment pairs are the same as those used in the P_{Σ_r}-TRAINED condition, detailed in Appendix E.1.

The subsequent preference elicitation procedure used to build the dataset of preferences for reward learning is identical to the procedure outlined in Section E.1, except subjects are asked "which path has the highest biggest possible final score" when generating preferences.

E.3 TRAINED-Control condition Interface details

When subjects interact with episodes from the grid-world domain, we display the "score so far", or the partial return, for the vehicle's path and explain this statistic in the same way as for the P_{Σ_r}-TRAINED condition. We do not train subjects to compute any segment statistic, nor do we instruct them on how to generate preferences like in the other conditions. During preference elicitation, we ask subjects "Which path do you prefer?", a QUESTION that does not seek to influence subjects towards any preference model. This preference elicitation procedure aims to be generally representative of standard approaches for collecting feedback for RLHF.

E.4 Testing subjects comprehension of the taught preference model

When teaching subjects to follow a specific preference model, we present them with two sets of six practice segment pairs and provide feedback on their preference labels for these pairs. We test the correlation between the subject's adherence to the taught preference model during the last six practice pairs and in the main preference elicitation portion of the study. We compute the Spearman correlation coefficient between the fraction of human preferences that agree with the noiseless version of the taught model in the last six practice segment pairs and in the fifty segment pairs shown during the main preference elicitation portion.

We are not able to perform this analysis for the P_{Σ_r}-TRAINED condition; the fraction of human preferences that agree with the noiseless version of the partial return preference model in the last six practice segment pairs remains constant for all subjects and therefore the Spearman correlation coefficient is undefined. For the P_{regret}-TRAINED condition, we compute a Spearman correlation coefficient of -0.137 with $p = 0.706$. We suspect that the high p-value is a result of the small sample size of only 10 subjects.

E.5 Surveying subject agreement of the taught preference model

During the post-study task-comprehension survey, we assess subjects' personal agreement with the taught preference model. For the P_{Σ_r}-TRAINED condition, we ask, "We told you that the better path is always the one with the higher SCORE SO FAR. How often did you agree with this?" For the P_{regret}-TRAINED condition, we ask, "We told you that the better path is always the one with the higher BIGGEST POSSIBLE FINAL SCORE. How often did you agree with this?" Responses are given on a 7-point Likert scale, where 1 indicates "always disagreed" and 7 indicates "always agreed." The mean response for the P_{Σ_r}-TRAINED condition was 4.2 with a variance of 1.56, while the P_{regret}-TRAINED condition had a mean response of 6.3 with a variance of 1.61. These results suggest that subjects personally aligned more with the regret-based labeling of segment pairs than with the partial return-based approach.

For both the P_{Σ_r}-TRAINED and P_{regret}-TRAINED conditions, we also asked subjects "How helpful were our explanations on why one path was better than another path for your own decision making?" The mean response for the P_{Σ_r}-TRAINED condition was 5.8 with a variance of 2.36, while the P_{regret}-TRAINED condition had a mean response of 6.4 with a variance of 0.84. We interpret these results as general satisfaction with our protocol for teaching subjects about a specific preference model.

E.6 Number of preferences per collected dataset

We collected 500 preferences for each condition outlined in the TRAINED experiment. Because we discard samples where a subject chose "Can't Tell" instead of a preference, each dataset contains a different number of preferences indicated in Table 2.

F QUESTION Experiment Interface details

The conditions in the QUESTION experiment only differ in what question is asked during preference elicitation. The control condition used for the QUESTION experiment is identical to that used for the TRAINED experiment, detailed in Appendix E.3.

Table 2: The number of preferences in each preference dataset resulting from the TRAINED experiment.

Condition	Number of Preferences in Dataset
P_{Σ_r}-TRAINED	497
TRAINED-Control	473
P_{regret}-TRAINED	491

F.1 Choosing the preference elicitation question

The authors of this paper were asked to propose possible questions to ask subjects during preference elicitation that might influence their preferences toward either preference model. The first author selected the three most appealing options for each preference model, each author ranked these options in order of desirability, and then ranked-choice voting was employed to select the winner. For guiding human preferences towards the regret preference model the three options ranked by each author were "Which path shows better decision-making?", "Which path reflects better decision-making?, and "Which path is more likely to be taken by an expert?". For guiding human preferences towards the partial return preference model, the three options were "Which path would be better if the task ended after the path?", "Which path has better immediate outcomes?", and "Which path looks better, considering only exactly what happened during the path?". After ranking these questions by their likely ability to guide human preferences towards the regret or partial return preference models, the second question won 80% and 60% of the time respectively.

F.2 Number of preferences per collected dataset

Table 3 displays the number of preferences collected for each condition in the QUESTION experiment after discarding samples where a subject chose "Can't Tell" instead of a preference.

G Correlations between preferences and segment statistics

Recall that we compute the regret of a trajectory segment with deterministic transitions as follows: $regret_d(\sigma|\tilde{r}) = V^*_{\tilde{r}}(s^\sigma_0) - (\Sigma_\sigma \tilde{r} + V^*_{\tilde{r}}(s^\sigma_{|\sigma|}))$, where one of the 3 components of regret is partial return, $\Sigma_\sigma \tilde{r}$. We combine two components of $regret_{\mathrm{d}}(\sigma|r)$ to simplify analysis, introducing the following shorthand: $\Delta_\sigma V_{\tilde{r}} \triangleq V^*_{\tilde{r}}(s^\sigma_{|\sigma|}) - V^*_{\tilde{r}}(s^\sigma_0)$.

Table 3: The number of preferences in each preference dataset resulting from the QUESTION experiment.

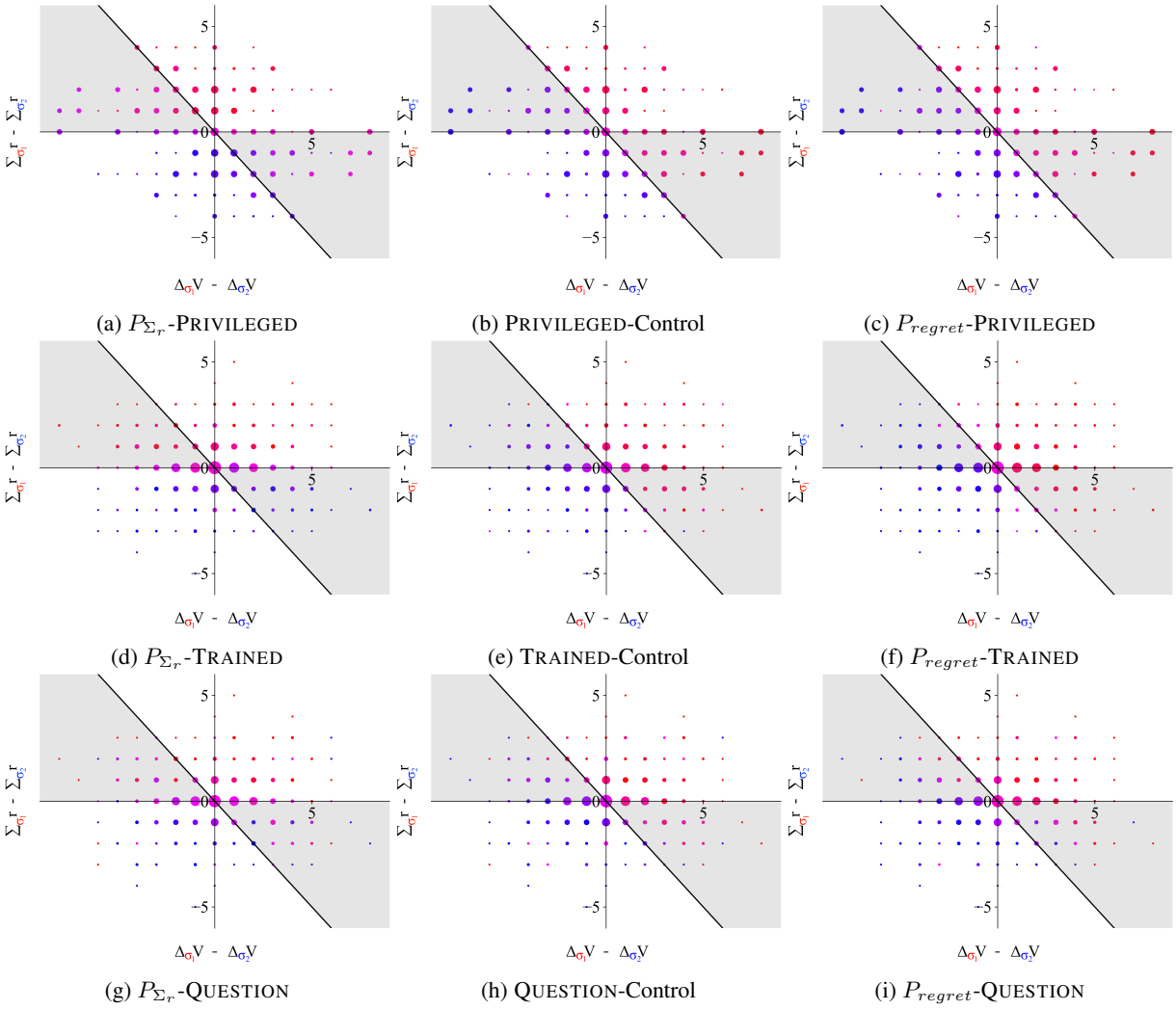
Condition	Number of Preferences in Dataset
P_{Σ_r}-QUESTION	437
QUESTION-Control	434
P_{regret}-QUESTION	442

The change in state value, $\Delta_\sigma V_{\tilde{r}}$, should have a greater effect on human preferences that are more aligned with the regret preference model, and partial return should have a greater effect on human preferences more aligned with the partial return preference model. The datasets of preferences are visualized in Figure 19. Note that on the diagonal line in Figure 19, $regret_{\mathrm{d}}(\sigma_2|r) = regret_{\mathrm{d}}(\sigma_1|r)$, making the P_{regret} preference model indifferent.

Figure 19 shows that, when influencing human preferences towards the regret preference model in the PRIVILEGED and TRAINED experiments, $\Delta_\sigma V_{\tilde{r}}$ has influence on the resulting preference dataset independent of partial return. This is

evident for the P_{regret}-PRIVILEGED and P_{regret}-TRAINED condition's dataset plots when focusing only on points at a chosen y-axis value; if the colors along the corresponding horizontal line reddens as the x-axis value increases, then $\Delta_\sigma V_{\tilde{r}}$ appears to have independent influence. Visually, $\Delta_\sigma V_{\tilde{r}}$ also exhibits independent influence on the preference datasets from the control condition for all experiments. When influencing human preferences towards the partial return preference model in the PRIVILEGED and TRAINED experiments, $\Delta_\sigma V_{\tilde{r}}$, has significantly less influence on the resulting preference dataset as evident by the x-axis–rather than the diagonal line–partitioning most red and blue points in the P_{Σ_r}-PRIVILEGED and P_{Σ_r}-TRAINED condition's dataset plots.

Visual inspection leads us to conclude that $\Delta_\sigma V_{\tilde{r}}$ independently influences human preferences for all conditions in the QUESTION experiment, indicating that regardless of the preference elicitation question, the change in state value is still correlated with how subjects label preferences.



(a) P_{Σ_r}-PRIVILEGED (b) PRIVILEGED-Control (c) P_{regret}-PRIVILEGED

(d) P_{Σ_r}-TRAINED (e) TRAINED-Control (f) P_{regret}-TRAINED

(g) P_{Σ_r}-QUESTION (h) QUESTION-Control (i) P_{regret}-QUESTION

Figure 19: Proportions of subjects preferring each segment in a pair, plotted by the difference in segments' change in state values (x-axis) and partial returns (y-axis). The areas of the circles are proportional to the number of segment pairs at that point. The proportionality is consistent across across all plots for the TRAINED and QUESTION experiments, and separately for the PRIVILEGED experiments. The diagonal line indicates points of indifference for P_{regret}, while indifference points for P_{Σ_r} are on the x-axis. The shaded gray area highlights where the partial return and regret preference models disagree, each preferring a different segment. To visually assess which preference model better fits the data: if subjects used the partial return preference model to generate preferences, the color gradient would be orthogonal to the x-axis. Conversely, if they followed the regret preference model, the gradient would be orthogonal to the diagonal line, as regret here is $x + y$.